

Preprocessing Project

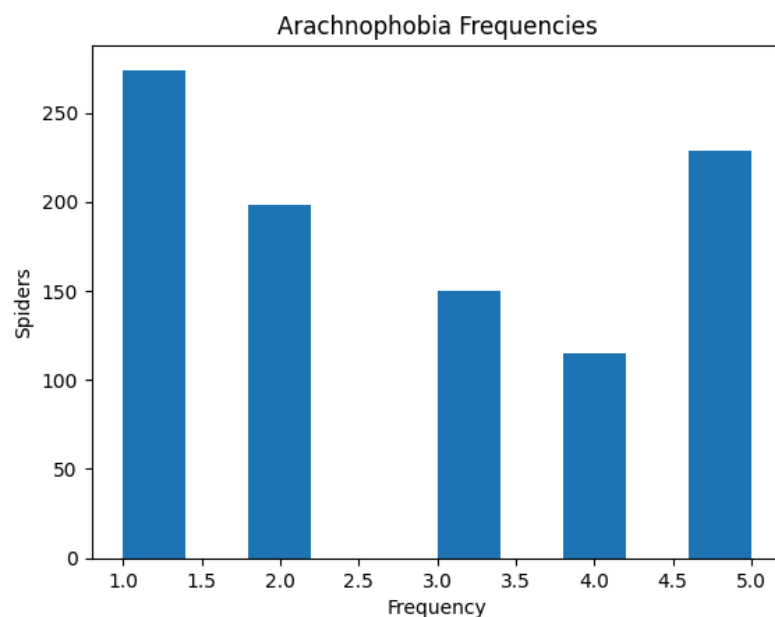
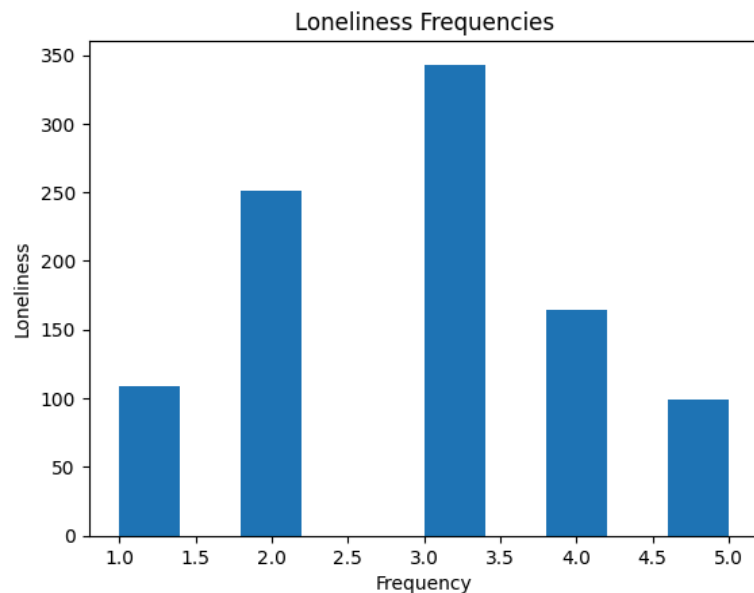
Initial Data Exploration

The data had categorical columns that could be ordinal and one hot encoded. One of the columns that lent well to ordinal encoding was internet usage. The two that made more sense one-hot encoded were village - town and gender. Initially, there were 1010 rows and 16 columns. Our independent variable was the Loneliness score. There is still some uncertainty with what each loneliness score means, though.

index	Unamed:0	Musical	Techno	Movies	History	Mathematics	Pets	Spirals	Loneliness	Parents' advice	Internet usage	Finances	Age	Siblings	Gender	Village - town
0	0	5.0	1.0	5.0	1.0	3.0	4.0	1.0	3.0	4.0	few hours a day	3.0	20.0	1.0	female	village
1	1	4.0	1.0	5.0	1.0	5.0	5.0	1.0	2.0	2.0	few hours a day	3.0	19.0	2.0	female	city
2	2	5.0	1.0	5.0	1.0	5.0	5.0	1.0	5.0	3.0	few hours a day	2.0	20.0	2.0	female	city
3	3	5.0	2.0	5.0	4.0	4.0	1.0	5.0	5.0	2.0	most of the day	2.0	22.0	1.0	female	city
4	4	5.0	2.0	5.0	3.0	2.0	1.0	1.0	3.0	3.0	few hours a day	4.0	20.0	1.0	female	Village

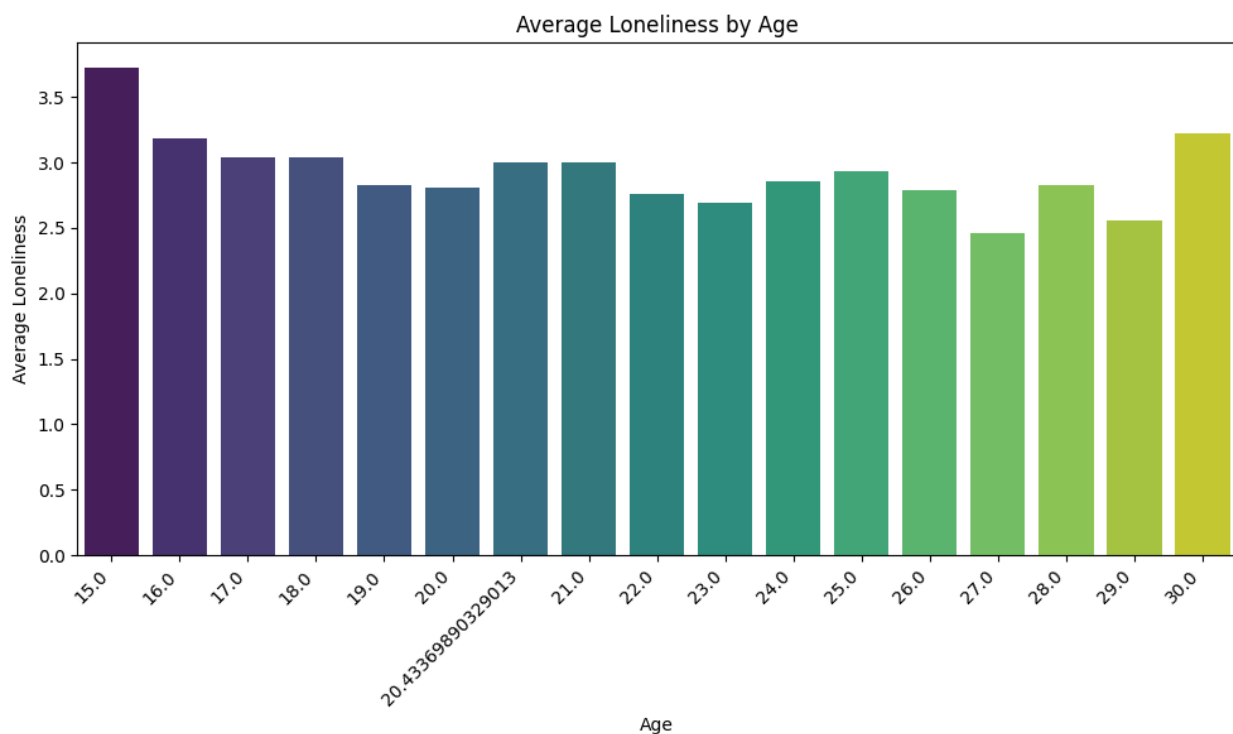
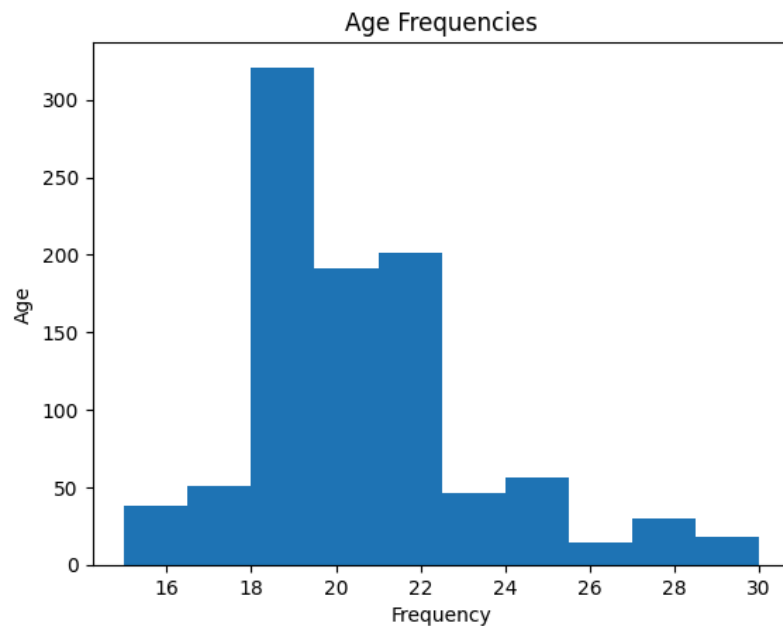
EDA (Exploratory Data Analysis)

Loneliness had a somewhat normal distribution. Most people fall somewhere in the middle with loneliness, where they don't feel completely lonely, but they still feel a sense of loneliness. None of the variables had a strong correlation with the independent variable, but out of all of the variables, fear of spiders and internet usage seems to have the strongest relationship with loneliness.

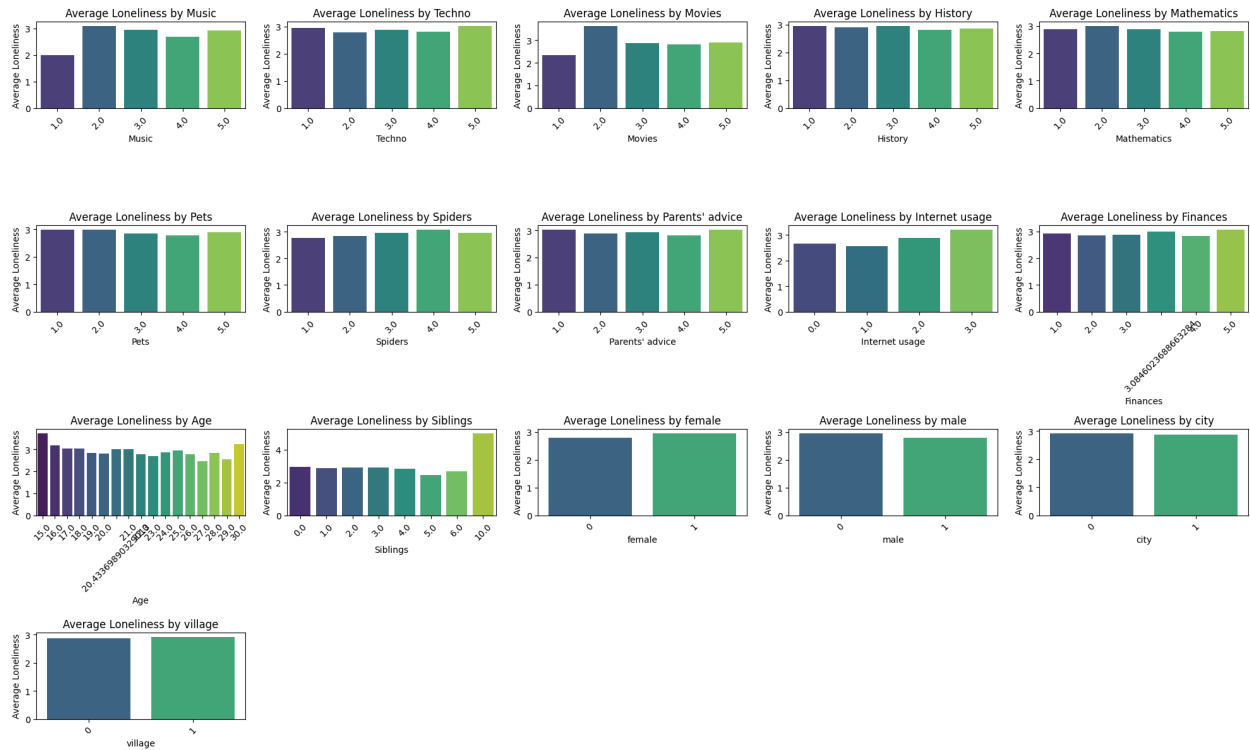


You might also notice how the frequency for those with arachnophobia are distributed. A majority of the participants in the study are one of the extremes when it comes to their fear of spiders. It could just be that the samples weren't large enough for a distribution. The data for

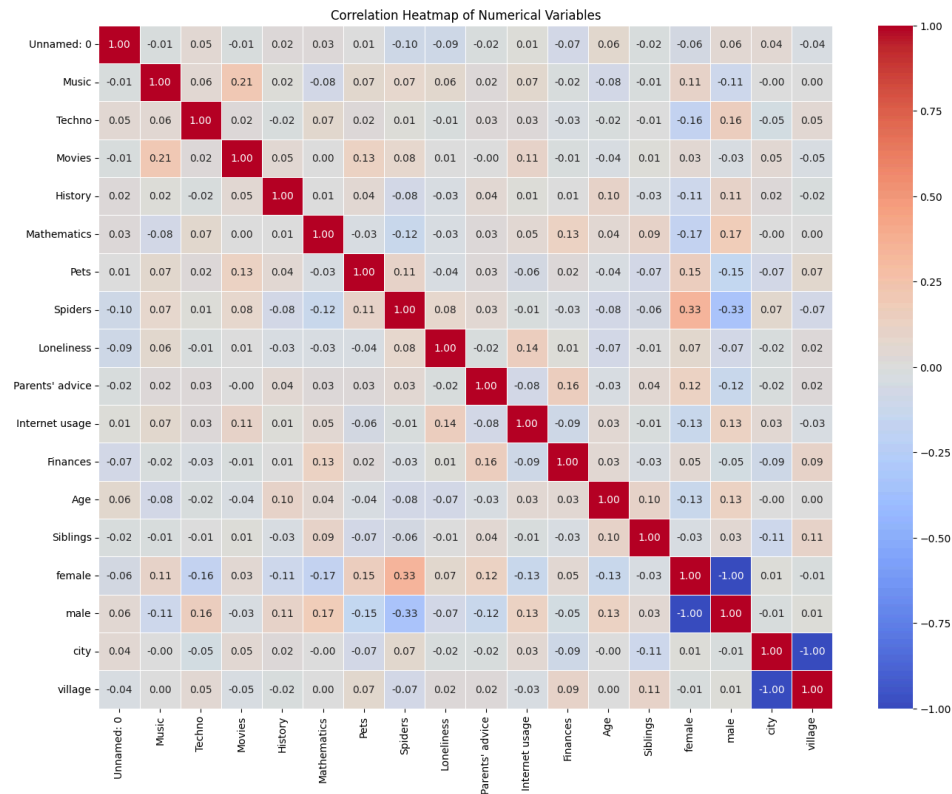
age was skewed a lot, due to the samples for each of the age groups being completely different sizes. There were very few fifteen year olds in this study, which caused the average for loneliness to be a lot higher for that younger age group. We realized age wasn't a very good predictive variable for loneliness, as the samples weren't taken very evenly.



The figure below shows all of the other variables and their correlation with the independent variable.



Finally, the image below shows a heatmap of the different variables and their relationships with each other. Internet usage and spiders seemed to have the strongest correlations with loneliness.



Most variables show little to no correlation to Loneliness the ones that were correlated the most were spiders and internet usage. All the rest had correlation of less than .07.

Data preprocessing

There were a couple of null values in the table but not more than the limit for some of them we decided to propagate with the average such as age and financial data. For others we decided against that. There were so few values that we decided that it wasn't worth it to try to propagate them.

Music	0.002970
Techno	0.006931
Movies	0.005941
History	0.001980
Mathematics	0.002970
Pets	0.003960
Spiders	0.004950
Loneliness	0.000990
Parents' advice	0.001980
Internet usage	0.000000
Finances	0.002970
Age	0.006931
Siblings	0.005941
Gender	0.005941
Village - town	0.003960

*Not scaled to 100

Before

of rows: 1010

of columns: 16

After

of rows: 961

of columns: 16

Variable Name	Variable Type	Encoding Method	Justification	Resulting Columns
gender	Nominal	One-Hot	No inherent order	2 columns
Village - town	Nominal	One-Hot	No inherent order	2 columns
Internet usage	Ordinal	Ordinal	There was an inherent order	1 column

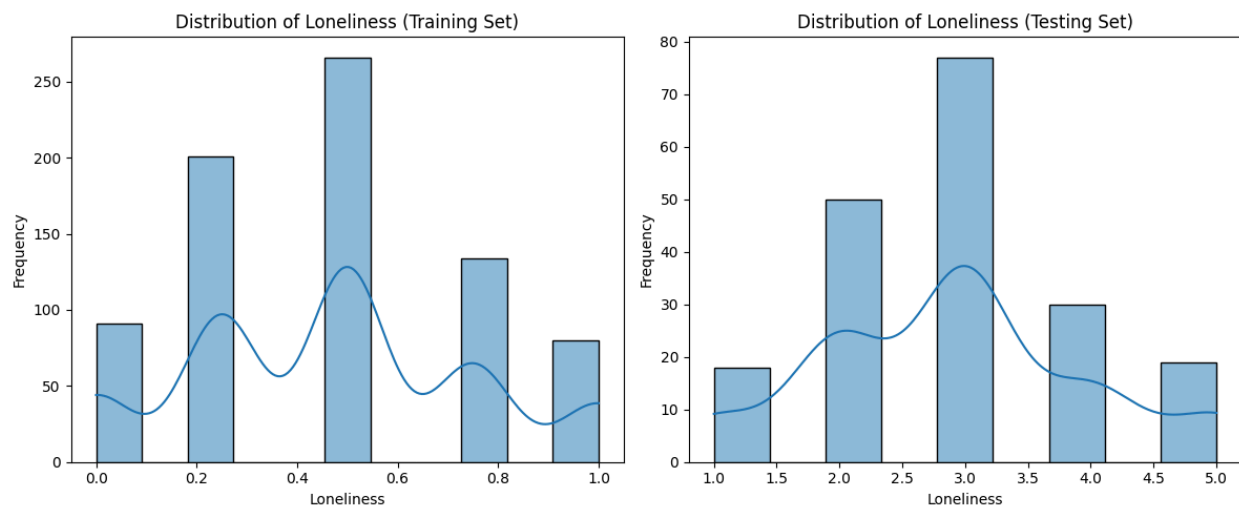
Training/Testing split

We decided to dedicate eighty percent of the data towards training the model and leave twenty percent for testing. We felt like there was plenty of data to be used here, and so having eighty percent of the data for training wasn't a huge deal, as we would still have 194 data points to use for testing.

Dataset	Samples	Features	% of Total
Training	772	16	80%
Testing	194	16	20%

The following image shows the two different distributions. On the left, you have the training set, and on the right you have the testing set. There is more of a distribution in the training set, but that is due to the larger sample used for training. Testing had a much smaller sample. The

testing distribution was pretty close to the training distribution with the amount of data it was given.



Feature Scaling

To train the model, the data was standardized. This meant that the training data was used as the fit and that fit was also applied to the testing data.

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

```
xfit = StandardScaler().fit(X_train)
yfit = StandardScaler().fit(y_train.values.reshape(-1, 1))
```

```
X_train = xfit.transform(X_train)
X_test = xfit.transform(X_test)
y_train = yfit.transform(y_train.values.reshape(-1, 1))
y_test = yfit.transform(y_test.values.reshape(-1, 1))
```

Summary

Because we had a lot of data and not much of it correlated we decided to go with random forest regression. This allowed us to use the not very closely correlated set and predict slightly better than we would have been able to with multivariate linear regression. We split the gender column into two different columns labeled male and female. We also split the Village - town variable into city and village features. We knew how we wanted to use the categorical columns upon the initial loading of the data, but further experimentation was required for us to find the missing values and fill or drop them.

References

All of the code is in this google collab sheet

<https://colab.research.google.com/drive/1uvonMtCjpMsjxHOWjv2JwBgprbXLoHpi?usp=sharing>